

1.Introduction

- **Project Title:** Customer Care Registry
- **Team Members:**

K.Jyothi swaroop(Team leader)

K.Bhu Mahesh

N.Mohith

Project Introduction: Customer Care Registry is a web-based Customer Relationship Management (CRM) application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The system is designed to simplify customer service operations by providing a centralized platform for managing customer information, complaints, service requests, follow-ups, and reports.

The application enables administrators and customer care executives to efficiently register customers, assign complaints, update complaint status, monitor customer interactions, and generate reports. Secure authentication using JWT ensures that only authorized users can access the system.

2. Project Overview

Purpose

The main objective of the Customer Care Registry is to improve customer service management by replacing manual record keeping with a digital system. The application allows organizations to track customer complaints, manage customer records, monitor issue resolution, and improve communication between customers and customer care executives.

Features

- User Registration
- Secure Login using JWT
- Customer Registration
- Customer Profile Management
- Complaint Registration
- Complaint Tracking
- Complaint Status Updates
- Dashboard with Statistics
- Customer Search
- Report Generation
- User Profile Management
- Secure Logout

3. Architecture

Frontend

The frontend of the application is developed using React.js. It provides an interactive and responsive user interface where users can register, login, manage customers, register complaints, and view reports.

Technologies Used

- React.js
- HTML5
- CSS3
- JavaScript (ES6)
- Bootstrap / Tailwind CSS
- Axios

Backend

The backend is developed using Node.js and Express.js. It handles all business logic, authentication, authorization, API routing, and communication with the database.

Responsibilities

- User Authentication
- JWT Token Generation
- Password Encryption
- CRUD Operations
- REST APIs
- Input Validation
- Error Handling

Database

MongoDB is used as the database to securely store customer records, complaint details, user information, reports, and application data.

Collections

- Users
- Customers
- Complaints
- Reports

4. Setup Instructions

Software Requirements

- Node.js
- MongoDB
- Visual Studio Code
- Git
- npm

Installation Steps

Clone the repository

```
git clone https://github.com/username/customer-care-registry.git
```

Install frontend dependencies

```
cd clientnpm install
```

Install backend dependencies

```
cd server  
npm install
```

Create a `.env` file inside the server folder.

Example

```
PORT=5000  
MONGO_URI=your_mongodb_connection_string  
JWT_SECRET=your_secret_key
```

5. Folder Structure

```
Customer-Care-Registry  
├── client  
│   ├── public  
│   ├── src  
│   │   ├── assets  
│   │   ├── components  
│   │   ├── context  
│   │   ├── pages  
│   │   ├── services  
│   │   ├── App.js  
│   │   ├── index.js  
│   └── server  
│       ├── config  
│       ├── controllers  
│       ├── middleware  
│       ├── models  
│       ├── routes  
│       ├── utils  
│       ├── app.js  
│       └── server.js  
└── package.json
```

6. Running the Application

Start Backend

```
cd server  
npm run dev
```

Start Frontend

```
cd client  
npm start
```

Frontend

<http://localhost:3000>

Backend

<http://localhost:5000>

7. API Documentation

Method	Endpoint	Description
POST	/api/auth/register	Register User
POST	/api/auth/login	Login User
GET	/api/customers	Get All Customers
POST	/api/customers	Add Customer
PUT	/api/customers/:id	Update Customer
DELETE	/api/customers/:id	Delete Customer
GET	/api/complaints	Get Complaints
POST	/api/complaints	Register Complaint
PUT	/api/complaints/:id	Update Complaint
DELETE	/api/complaints/:id	Delete Complaint

8. Authentication

The Customer Care Registry uses JSON Web Token (JWT) authentication to protect application resources.

Authentication Features

- User Registration
- Secure Login
- Password Encryption using bcrypt
- JWT Token Generation
- Protected Routes
- User Logout
- Session Validation

Only authenticated users can access customer and complaint management features.

9. User Interface

The application provides a responsive and user-friendly interface.

Screens included

- Login Page
- Registration Page
- Dashboard
- Customer Management
- Complaint Management
- Reports
- Profile Page

Insert screenshots of each page with captions such as:

Figure 1: Login Page

Figure 2: Dashboard

Figure 3: Compliant tracking

Figure 4: Profile

10. Testing

The application was tested to ensure all functionalities work correctly.

Testing Performed

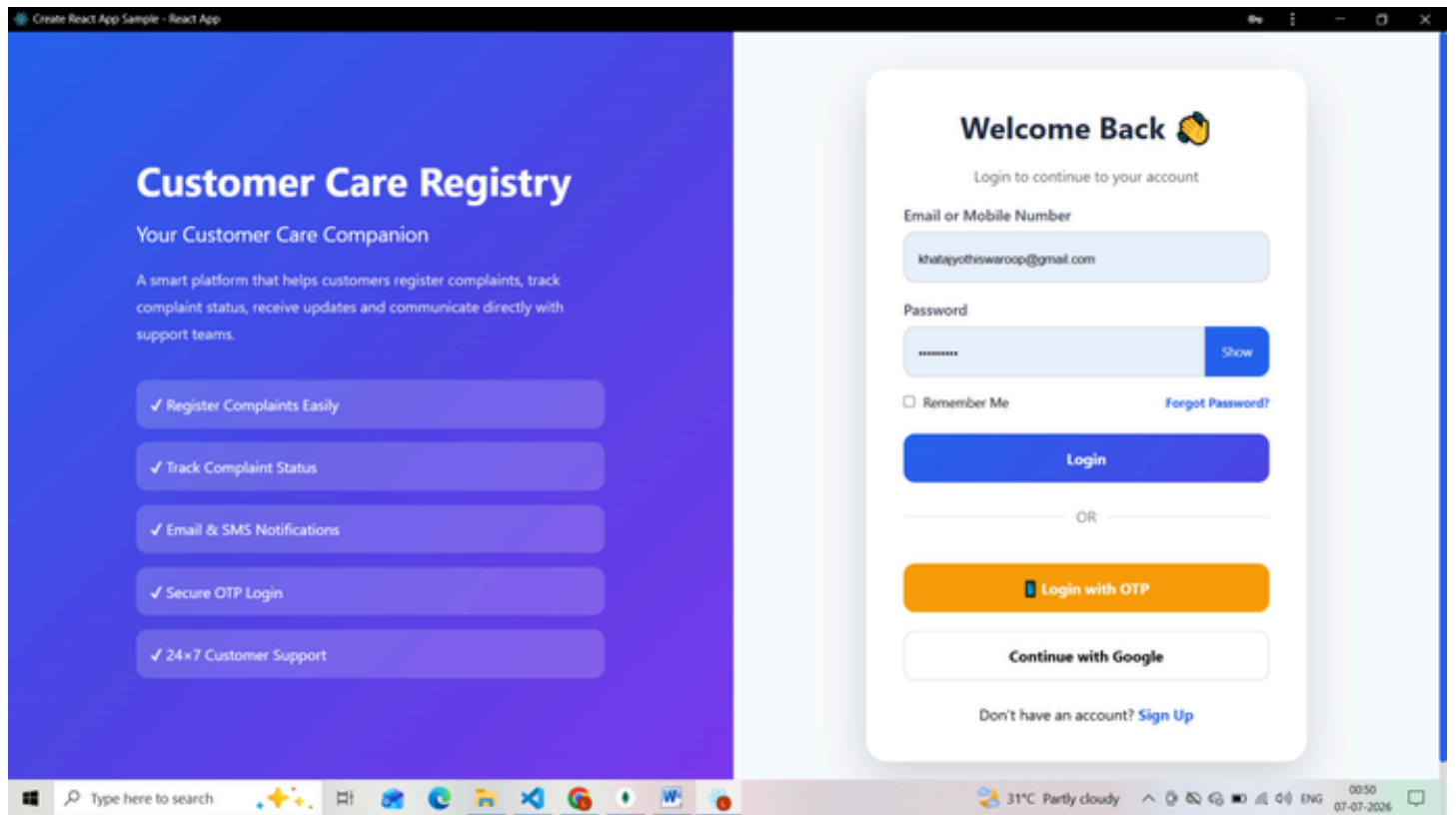
- User Registration Testing
- Login Authentication Testing
- Customer CRUD Testing
- Complaint CRUD Testing
- Dashboard Testing
- API Testing using Postman
- MongoDB Database Testing
- Form Validation Testing
- Responsive UI Testing
- Security Testing

All major modules were successfully tested and validated.

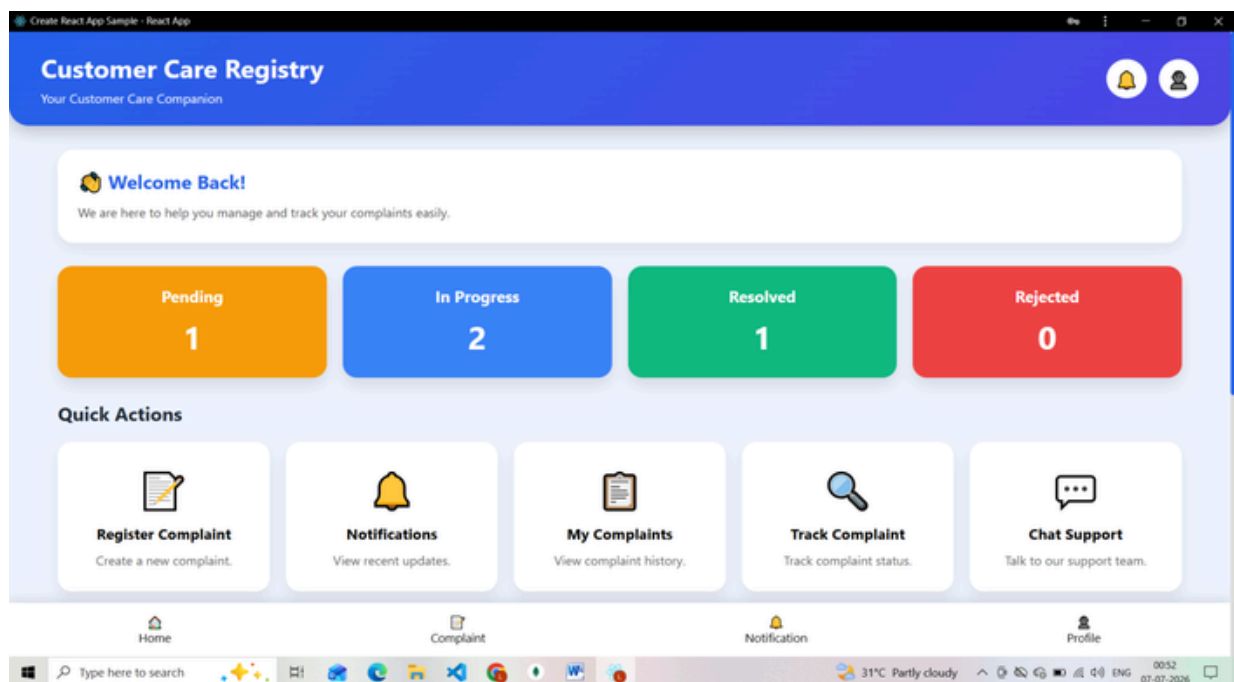
11. Screenshots / Demo

Include screenshots of the following application pages:

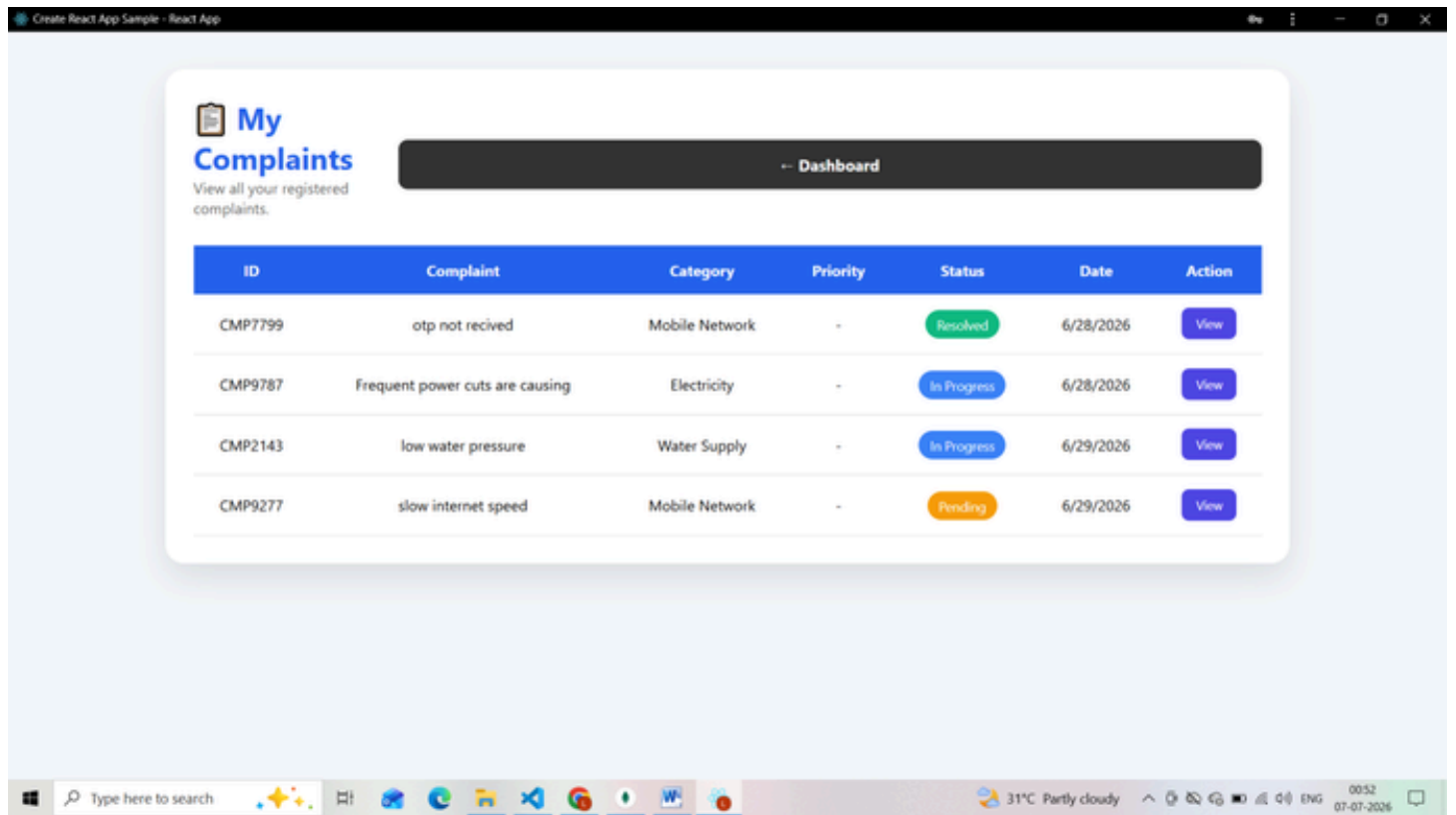
- Login Screen
- User Registration
- Dashboard
- Customer Management
- Complaint Management
- Reports
- Profile Page



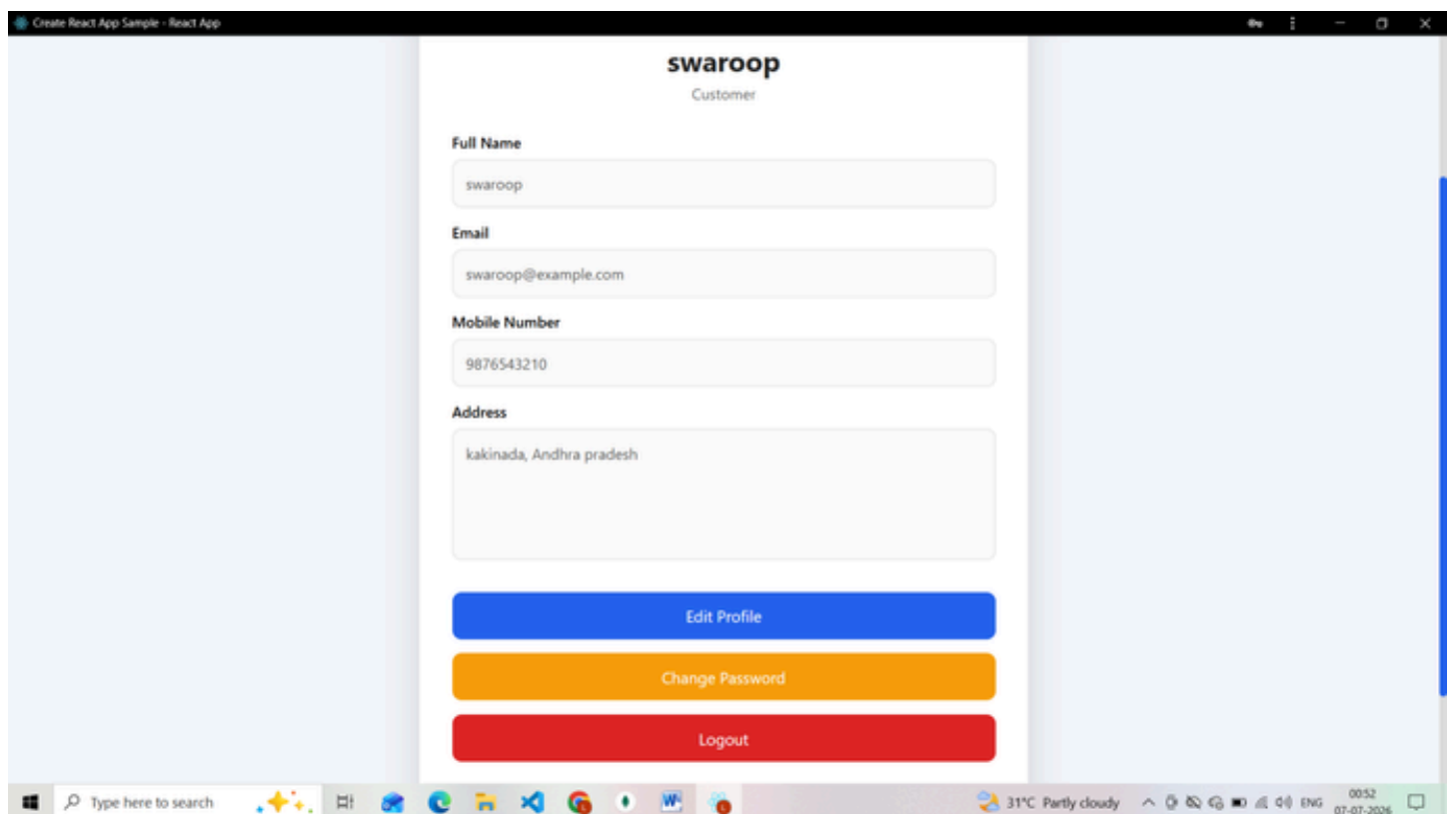
Fig(1)



Fig(2)



Fig(3)



Fig(4)

12. Known Issues

The current version has the following limitations:

- SMS alerts are not available.
- Advanced reporting can be improved.
- Mobile application is not available.
- Multi-language support is not implemented.

13. Future Enhancements

The following features can be added in future versions:

- Mobile Application
- SMS Notifications
- PDF Report Generation
- Excel Report Export
- Role-Based Access Control
- Real-Time Notifications
- Advanced Analytics Dashboard
- Cloud Deployment with CI/CD
- Multi-language Support